

Package ‘PhyloTraitDynamics’

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Title Companion Code for Phylogenetic Trait Dynamics

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Description Companion code for the article “Phylogenetic dynamics of MRCA ages and empirical moments of a Brownian trait”. Public functions assume inputs consistent with the article's assumptions.

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birth_death_brownian_simulate

Simulate Birth-Death Brownian Empirical Moments

Description

Simulates birth-death Brownian paths and records, on the requested time grid, the number of living lineages, the empirical mean of their traits, and their empirical variance.

Usage

```
birth_death_brownian_simulate(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  B,
  x0 = 0,
  seed = NULL,
  n_envelope = 4000,
  safety_factor = 1.05
)
```

Arguments

birth	Non-negative birth rate, either a numeric constant or a function of time.
death	Non-negative death rate, either a numeric constant or a function of time.
sigma2	Brownian variance parameter.
time_end, time_step	Time grid definition.
B	Number of simulated paths.
x0	Initial trait value.
seed	Optional random seed.
n_envelope	Number of points used by the thinning envelope.
safety_factor	Multiplicative safety factor for the thinning envelope.

Value

An object of class "birth_death_brownian_simulation" containing the time grid, matrices of empirical means, empirical variances and lineage counts, and a simulation summary.

empirical_mean_compute_variance

Variance of the Empirical Mean Under Birth-Death Brownian Dynamics

Description

Computes the theoretical variance of the empirical mean of Brownian traits, conditional on survival at each time.

Usage

```
empirical_mean_compute_variance(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  n_steps = 1200,
  tol_sing = 1e-07,
  li2_tol = 1e-12
)
```

Arguments

birth	Non-negative birth rate, either a numeric constant or a function of time.
death	Non-negative death rate, either a numeric constant or a function of time.
sigma2	Brownian variance parameter.
time_end, time_step	Time grid definition.
n_steps	Number of integration steps for numerical theory.
tol_sing, li2_tol	Numerical tolerances.

Value

A data frame with time, survival probability, empirical mean variance, and conditioning columns. The original script result is attached as attribute "original_result".

empirical_mean_plot_variance

Plot Variance of the Empirical Mean

Description

Plots Monte Carlo summaries and/or theoretical values for the variance of the empirical mean.

Usage

```
empirical_mean_plot_variance(
  simulation = NULL,
  theory = NULL,
  main = NULL,
  xlab = "time",
  ylab = "Variance of empirical mean"
)
```

Arguments

`simulation` Optional "empirical_mean_simulation" object.

`theory` Optional data frame returned by `empirical_mean_compute_variance()`.

`main, xlab, ylab` Base graphics labels.

Value

Invisibly returns a list containing `simulation`, `theory`, and `summary`.

`empirical_variance_compute_expectation`

Expected Empirical Variance Under Birth-Death Brownian Dynamics

Description

Computes the theoretical expectation of the empirical trait variance on a time grid for a birth-death process with Brownian trait evolution.

Usage

```
empirical_variance_compute_expectation(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  conditioning = c("none", "survival"),
  method = c("auto", "numeric", "closed_constant"),
  n_steps = 2000,
  tol = 1e-08,
  constant_tol = 1e-10,
  li2_tol = 1e-12
)
```

Arguments

`birth` Non-negative birth rate, either a numeric constant or a function of time.

`death` Non-negative death rate, either a numeric constant or a function of time.

`sigma2` Brownian variance parameter.

time_end, time_step	Time grid definition.
conditioning	Conditioning used for the expectation. "none" is unconditional and "survival" conditions on $N(t) > 0$.
method	Theory evaluation method. "auto" uses the closed constant-rate formula when the birth and death rates are constant on the time grid, and otherwise uses the numerical method. "closed_constant" requires constant birth and death rates.
n_steps	Number of integration steps for numerical theory.
tol, constant_tol, li2_tol	Numerical tolerances passed to the original implementation.

Value

A data frame with time, survival probability, expected empirical variance, conditioning, and method columns. The original script result is attached as attribute "original_result".

empirical_variance_plot_expectation
Plot Expected Empirical Variance

Description

Plots simulated empirical variance paths, Monte Carlo summaries, and/or the theoretical expectation on a common set of axes.

Usage

```
empirical_variance_plot_expectation(
  simulation = NULL,
  theory = NULL,
  show_paths = TRUE,
  conditioning = c("none", "survival"),
  max_paths = Inf,
  alpha = 0.05,
  main = NULL,
  xlab = "time",
  ylab = "Empirical variance"
)
```

Arguments

simulation	Optional "empirical_variance_simulation" object.
theory	Optional data frame returned by empirical_variance_compute_expectation().
show_paths	Logical; whether to draw individual simulated paths.
conditioning	Which expectation to plot.
max_paths	Maximum number of simulated paths to draw.
alpha	Transparency used for simulated paths.
main, xlab, ylab	Base graphics labels.

Value

Invisibly returns a list containing simulation, theory, and summary.

fixed_tree_compute_theoretical_summary

Theoretical Summary on a Fixed Tree

Description

Computes theoretical summaries of empirical mean variance and empirical variance for Brownian traits on a fixed phylogenetic tree.

Usage

```
fixed_tree_compute_theoretical_summary(
  tree,
  sigma2 = 1,
  time_end = NULL,
  time_step
)
```

Arguments

tree	A phylogenetic tree of class "phylo".
sigma2	Brownian variance parameter.
time_end	End of the time grid. If NULL, the maximum tree time is used.
time_step	Time step used to define the time grid.

Value

A data frame with time, lineage counts, empirical mean variance, empirical variance expectation, empirical variance quartiles, and empirical variance variance. The tree, sigma2, and original result are attached as attributes.

fixed_tree_plot_brownian_realization

Plot a Fixed-Tree Brownian Realization

Description

Plots the simulated Brownian realization together with the fixed tree using the original plotting routine.

Usage

```
fixed_tree_plot_brownian_realization(
  x,
  n_bands = 10,
  time_band_col = grDevices::adjustcolor("blue", alpha.f = 0.05),
  cex = 1
)
```

Arguments

x	A <code>fixed_tree_brownian_realization</code> object.
n_bands	Number of time bands shown by the original plotting routine.
time_band_col	Color used for time bands.
cex	Character expansion factor passed to base graphics parameters.

Value

Invisibly returns x.

`fixed_tree_plot_theoretical_summary`

Plot a Fixed-Tree Theoretical Summary

Description

Plots fixed-tree theoretical summaries for empirical mean variance and empirical variance, optionally with the tree.

Usage

```
fixed_tree_plot_theoretical_summary(
  x,
  tree = NULL,
  show_tree = TRUE,
  main = NULL,
  time_band_col = grDevices::adjustcolor("blue", alpha.f = 0.05),
  band_col = grDevices::adjustcolor("red", alpha.f = 0.18),
  expectation_col = "blue",
  median_col = "red",
  mean_variance_col = "darkgreen",
  cex = 1
)
```

Arguments

x	A <code>fixed_tree_theoretical_summary</code> object, as returned by <code>fixed_tree_compute_theoretical_summary</code> .
tree	Optional phylogenetic tree of class "phylo". If NULL, the tree attached to x is used when available.
show_tree	Logical; whether to show the tree panel.
main	Optional main title.
time_band_col	Color used for time bands.
band_col	Color used for the empirical-variance interquartile band.
expectation_col	Color used for the empirical-variance expectation.
median_col	Color used for the empirical-variance median and quartiles.
mean_variance_col	Color used for the empirical-mean variance.
cex	Character expansion factor passed to base graphics parameters.

Value

Invisibly returns x.

```
fixed_tree_simulate_brownian_realization
```

Simulate Brownian Traits on a Fixed Tree

Description

Simulates one Brownian trait realization on a fixed phylogenetic tree and records empirical mean and empirical variance through time.

Usage

```
fixed_tree_simulate_brownian_realization(
  tree,
  sigma2 = 1,
  time_step,
  seed = NULL
)
```

Arguments

tree	A phylogenetic tree of class "phylo".
sigma2	Brownian variance parameter.
time_step	Time step used for the simulation output.
seed	Optional random seed.

Value

An object of class "fixed_tree_brownian_realization" containing the input tree, simulation data frame, time series, parameters, and seed.

```
mrca_age_compute_distribution
```

MRCA Age Distribution at One Time

Description

Computes the distribution and summary statistics of the most recent common ancestor age at a fixed time under a birth-death process.

Usage

```
mrca_age_compute_distribution(
  birth,
  death,
  time,
  time_step = NULL,
  probs = c(0.25, 0.5, 0.75),
  n_internal = NULL
)
```

Arguments

birth	Non-negative birth rate, either a numeric constant or a function of time.
death	Non-negative death rate, either a numeric constant or a function of time.
time	Observation time.
time_step	Internal time step. If NULL, a default step is derived from time.
probs	Quantile probabilities to report.
n_internal	Optional number of internal subintervals; when supplied it overrides time_step.

Value

An object of class "mrca_age_distribution" with distribution grid, summary statistics, requested probabilities, and the original result.

mrca_age_compute_dynamics

MRCA Age Distribution Dynamics

Description

Computes MRCA age distribution summaries over a time grid under a birth-death process.

Usage

```
mrca_age_compute_dynamics(
  birth,
  death,
  time_end,
  time_step,
  probs = c(0.25, 0.5, 0.75),
  n_internal_per_step = 10L,
  include_zero = TRUE
)
```

Arguments

birth	Non-negative birth rate, either a numeric constant or a function of time.
death	Non-negative death rate, either a numeric constant or a function of time.
time_end, time_step	Time grid definition.
probs	Quantile probabilities to report.
n_internal_per_step	Number of internal integration substeps per output time step.
include_zero	Logical; whether to include time zero in the original dynamics computation.

Value

An object of class "mrca_age_dynamic" containing the summary data frame, distribution grids, probabilities, and original call.

mrca_age_plot_dynamics

Plot MRCA Age Distribution Dynamics

Description

Plots MRCA age summaries over time, including median, optional interquartile band, and optional expectation.

Usage

```
mrca_age_plot_dynamics(
  x,
  show_iqr = TRUE,
  show_expectation = TRUE,
  main = NA,
  xlab = "Time",
  ylab = "MRCA age",
  add_legend = TRUE,
  ...
)
```

Arguments

x	An "mrca_age_dynamic" object.
show_iqr	Logical; whether to draw the interquartile band.
show_expectation	Logical; whether to draw the expectation curve.
main, xlab, ylab	Base graphics labels.
add_legend	Logical; whether to draw the plot legend.
...	Additional arguments passed to graphics::plot() .

Value

Invisibly returns x.

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